

BLUESTOCK MUTUAL FUND ANALYTICS

Final Project Report

End-to-End Data Engineering, Financial Analytics & Interactive Dashboard

Project Area	Mutual Fund Analytics
Selected Schemes	40
NAV Observations	46,000
Investor Transactions	32,778
Primary Tools	Python, Pandas, NumPy, SQL, SQLite, Power BI

This report documents the complete Bluestock Mutual Fund Analytics capstone, covering data ingestion, cleaning, validation, structured analysis, financial metrics, investor analytics, interactive dashboards and fund recommendation logic.

1. Executive Summary

The project builds an end-to-end mutual-fund analytics solution around a selected universe of 40 schemes. The supplied data includes 46,000 NAV observations, 32,778 investor transactions, 322 portfolio holdings and 8,050 benchmark observations.

At the latest AUM observation (31 Dec 2025), the selected fund-house series totals 62.74 lakh crore. The latest SIP inflow observation is ■31,002 crore and the latest folio observation is 26.12 crore.

The technical solution separates ingestion, transformation, validation, analytics and presentation. Python provides the processing and quantitative layer; SQL/SQLite supports structured storage; Power BI provides the interactive reporting layer.

Major outcomes

- Reusable ETL pipeline with logging, error handling, cleaning and validation.
- NAV-specific date processing, including daily reindexing and forward filling across weekends/holidays.
- Performance analysis covering CAGR, volatility, Sharpe Ratio, Beta, historical VaR and drawdown.
- Four-page Power BI dashboard plus fund-level NAV drill-through.
- Transparent risk-profile-based fund recommendation logic.

2. Objectives and Solution Architecture

- Automate mutual-fund data ingestion and preparation.
- Create reliable, analysis-ready financial datasets.
- Store and query data through a relational database.
- Perform EDA and financial performance analysis.
- Benchmark fund behaviour against market indices.
- Analyze SIP, AUM, folio and investor transaction trends.
- Deliver an interactive Power BI reporting experience.
- Rank funds using a transparent quantitative recommender.

Architecture

Raw CSV / API → Python ingestion → cleaning & validation → processed data → SQLite / SQL → EDA & performance metrics → recommender → Power BI → final report.

Layer	Technology	Role
Ingestion	Python / MFAPI / CSV	Acquire source data
Transformation	Pandas / NumPy	Clean and standardize data
Storage	SQLite / SQL	Relational storage and queries
Analytics	Python / Jupyter	EDA and financial metrics
Recommendation	Python	Risk-aware fund ranking
Visualization	Power BI	Interactive dashboards

3. Data Sources and Dataset Inventory

The project integrates complementary datasets so that scheme-level, industry-level, investor-level and benchmark questions can be analyzed within one solution.

Dataset	Rows	Use
Fund Master	40	Scheme metadata, category, plan, benchmark and risk
NAV History	46,000	Historical NAV and returns
AUM by Fund House	90	AUM trend and fund-house comparison
Monthly SIP Inflows	48	SIP trend and growth
Category Inflows	144	Category-level flow analysis
Industry Folios	21	Investor folio trend
Scheme Performance	40	Return, risk and performance metrics
Investor Transactions	32,778	Investor behaviour and geography
Portfolio Holdings	322	Underlying stock and sector exposure
Benchmark Indices	8,050	Market benchmark comparison

The selected scheme universe contains 40 schemes, while the supporting datasets provide wider industry and market context.

4. ETL Pipeline and Data Quality

The ETL layer is designed to make data preparation repeatable and auditable rather than dependent on manual spreadsheet operations.

Extraction

- Loads source CSVs from the raw-data layer.
- Supports dedicated live NAV retrieval.
- Checks for missing, empty or inaccessible source files.

Transformation

- Standardizes column names.
- Removes exact duplicates.
- Handles generic missing numeric values.
- Converts dates and financial values to appropriate data types.

NAV-specific handling

- Validates AMFI code, date and NAV columns.
- Removes duplicate AMFI code + date observations.
- Reindexes each scheme to a complete daily calendar.
- Forward-fills NAV across weekends and holidays.
- Rejects non-positive NAV values.

Validation

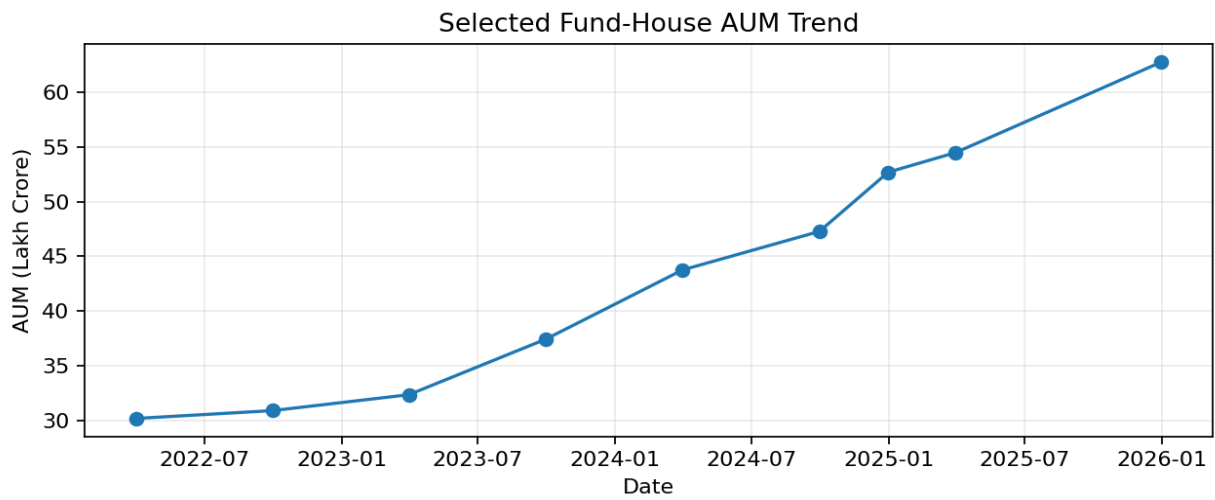
- Checks empty datasets, missing columns, duplicates, missing values and infinite numeric values.
- Validates NAV positivity and AMFI code + date uniqueness.

The implementation uses project-relative paths through pathlib instead of hard-coded machine-specific paths.

5. Exploratory Data Analysis

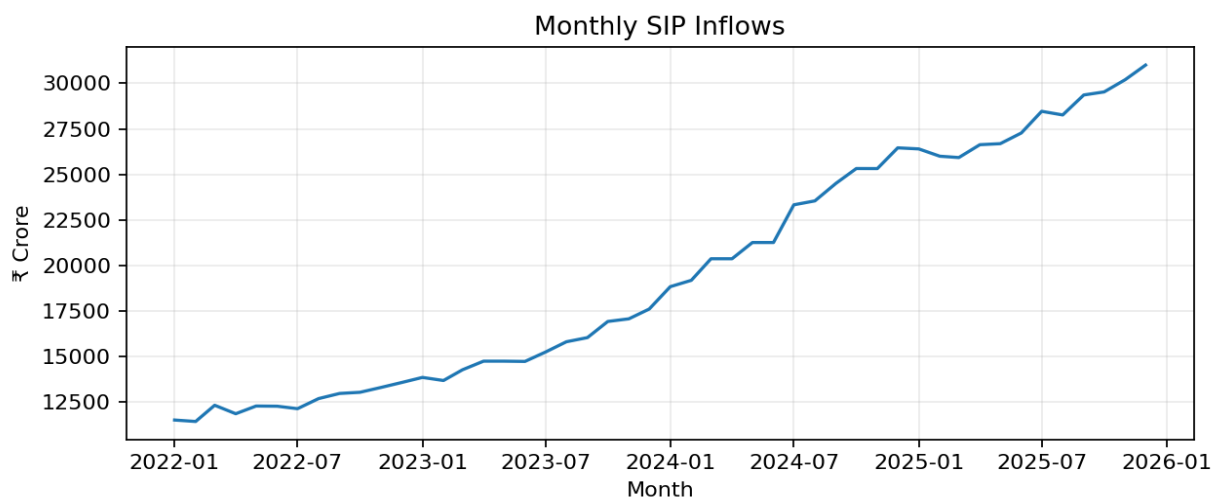
EDA establishes the scale and behaviour of the dataset before financial metrics and dashboard interpretation.

AUM trend



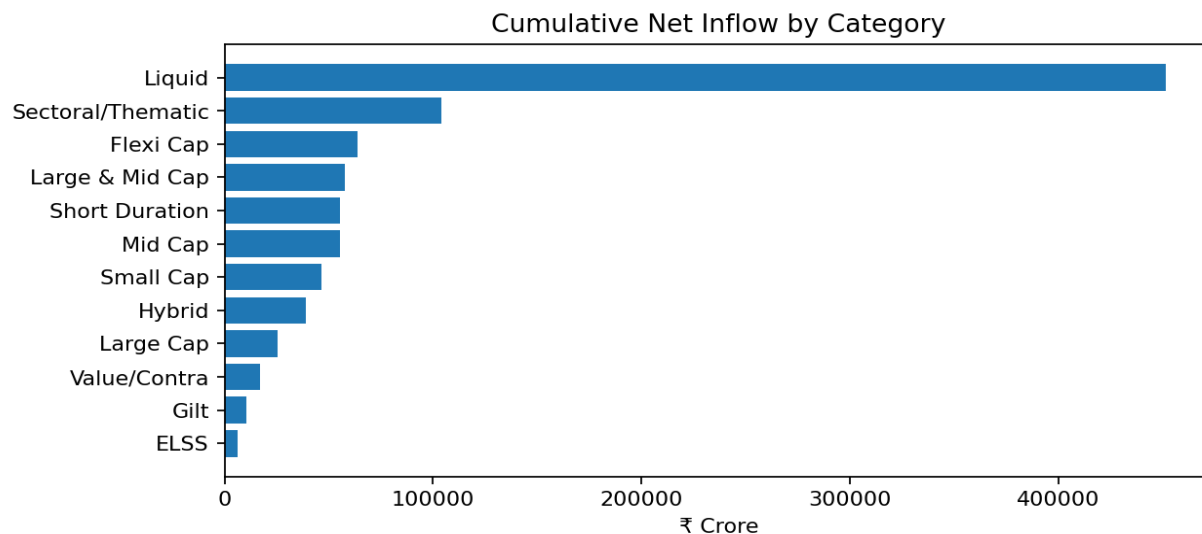
The selected fund-house AUM series rises over the observation period and reaches 62.74 lakh crore at the latest point.

SIP trend



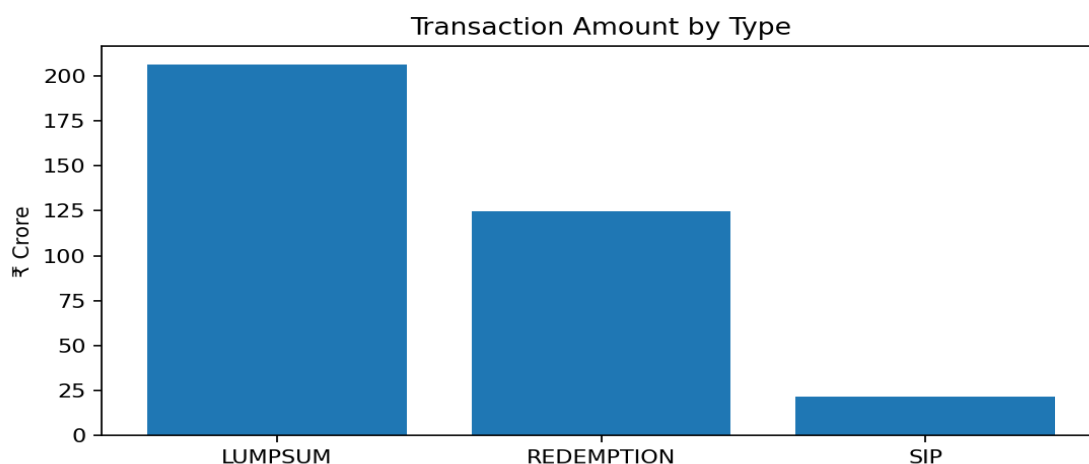
Monthly SIP inflows rise from ■11,517 crore in the first observation to ■31,002 crore in the latest observation.

6. Category and Investor Analytics



Liquid records the highest cumulative net inflow in the supplied category-flow dataset at approximately ■451,275 crore.

Investor transactions



The transaction dataset contains 32,778 records with total transaction value of approximately ■352 crore. Lumpsum transactions form the largest transaction-value component in the supplied sample.

7. Performance Metrics

The performance layer evaluates both return and risk so that a fund is not judged on raw return alone.

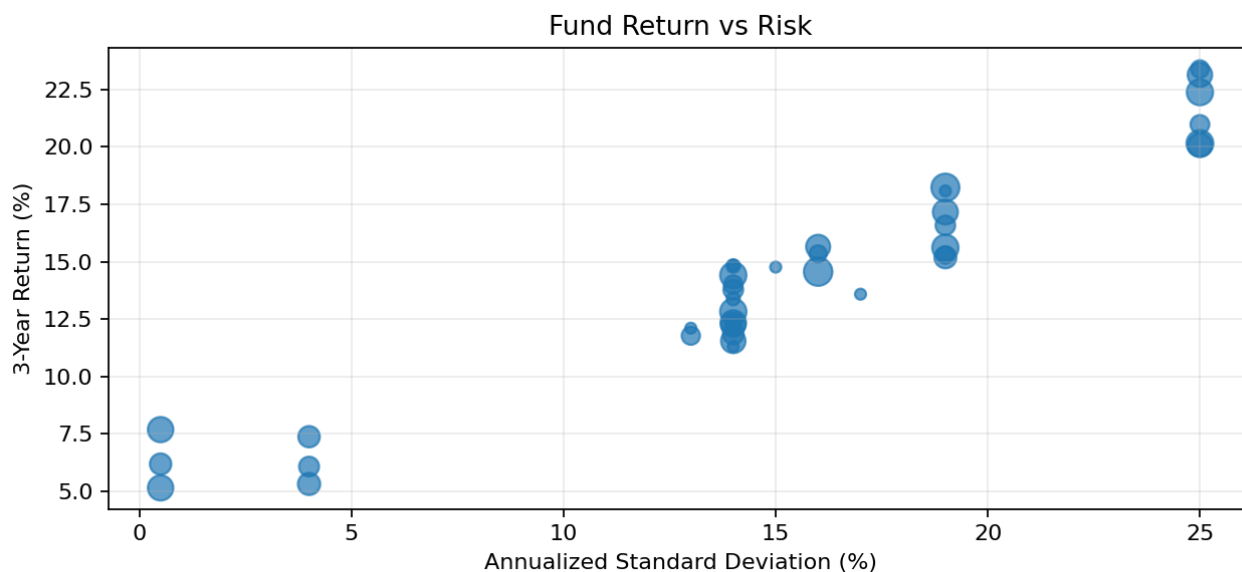
Metric	Purpose
Daily Return	Period-to-period NAV percentage change
CAGR	Compounded annual growth over 1Y, 3Y and 5Y
Volatility	Annualized standard deviation using 252 trading days
Sharpe Ratio	Risk-adjusted return using 6.5% risk-free rate
Beta	Sensitivity to NIFTY 50
Historical VaR (95%)	Historical downside threshold
Maximum Drawdown	Largest peak-to-trough decline

Results from the supplied scheme-performance table

Measure	Result
Average 1-year return	14.38%
Average 3-year return	14.09%
Average 5-year return	14.52%
Average annualized volatility	14.96%
Average Sharpe Ratio	1.36
Average Beta	0.87
Average maximum drawdown	-19.20%

Highest 3-year return: SBI Small Cap Fund - Regular Plan - Growth at 23.39%. Highest Sharpe Ratio: ICICI Pru Liquid Fund - Regular - Growth at 7.68. Highest alpha: HDFC Short Term Debt Fund - Regular - Growth at 1.98%.

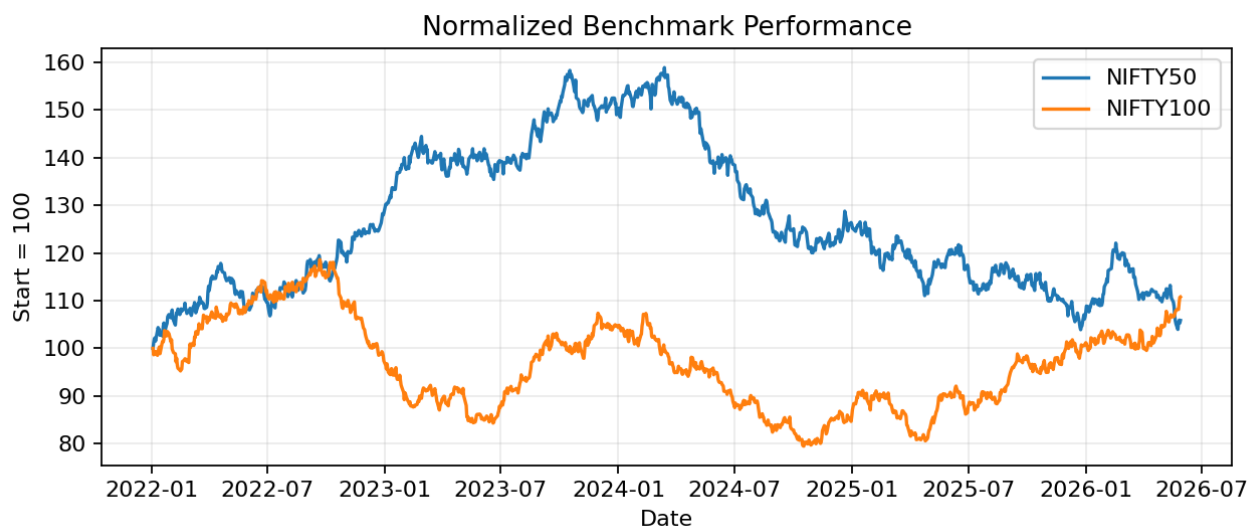
8. Risk–Return Analysis



The scatter plot compares 3-year return with annualized standard deviation; bubble size represents scheme AUM. This makes the return-risk trade-off visible and supports the fund scorecard used in Power BI.

Category-level results show that Small Cap has the highest average 3-year return among the supplied categories, while Liquid, Gilt and Short Duration categories have lower average returns and generally different risk characteristics.

9. Benchmark Analysis



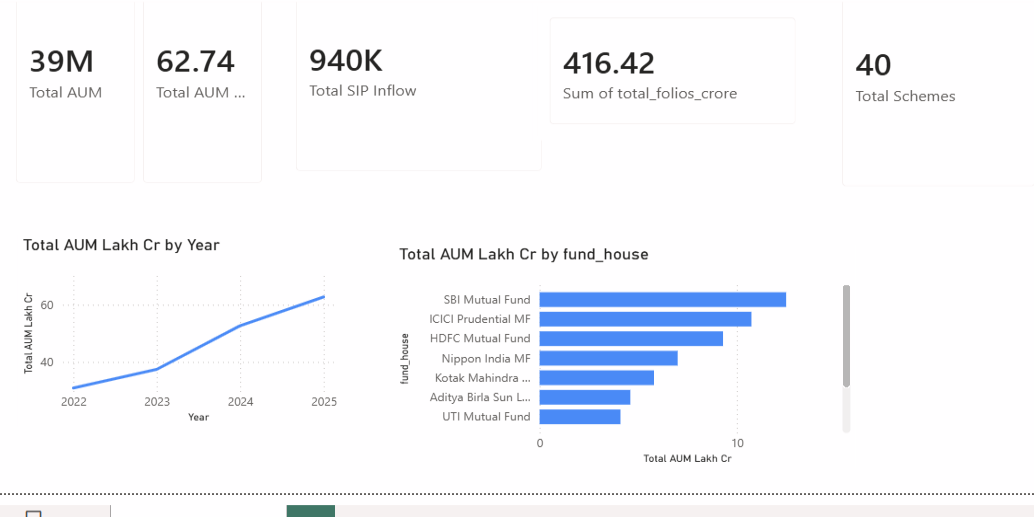
The benchmark layer contains seven market indices. Normalization to a common starting value of 100 allows NIFTY 50 and NIFTY 100 to be compared with fund NAV behaviour without confusing absolute index levels with NAV values.

Benchmark coverage includes NIFTY 50, NIFTY 100, NIFTY Midcap 150, BSE Smallcap, NIFTY 500, CRISIL Liquid and CRISIL Gilt.

10. Power BI – Industry Overview

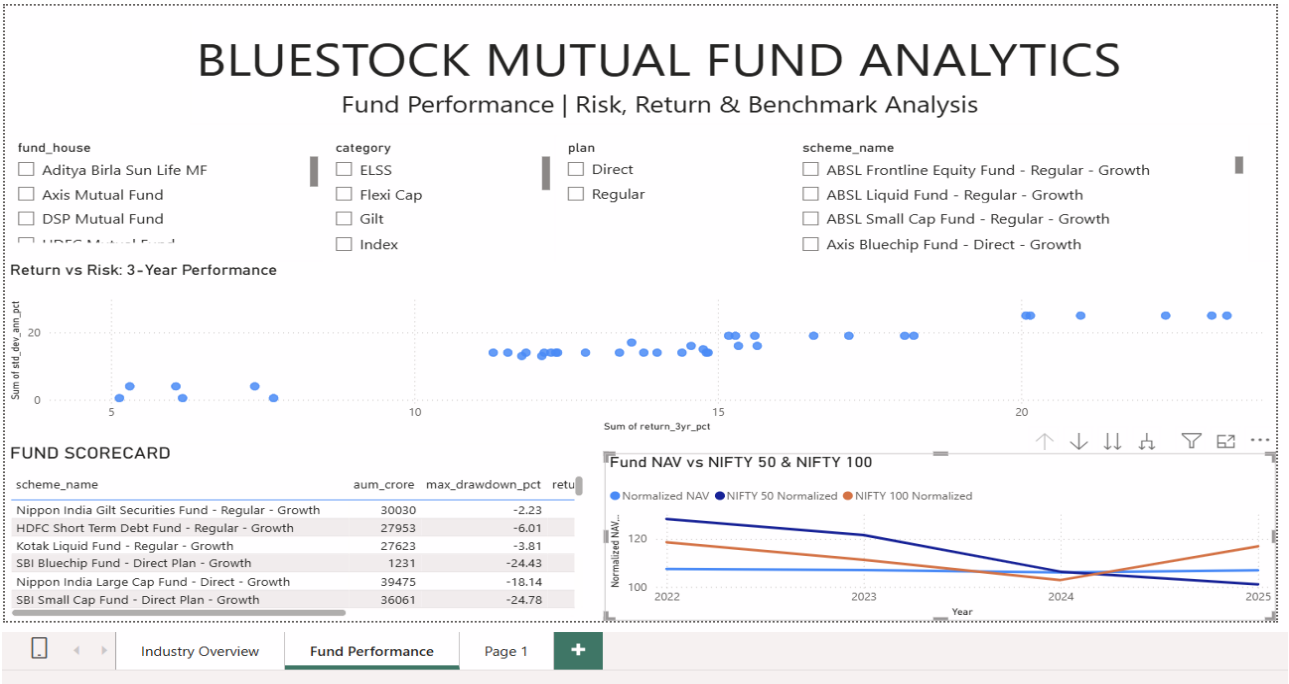
BLUESTOCK MUTUAL FUND ANALYTICS

Industry Overview | 2022–2025



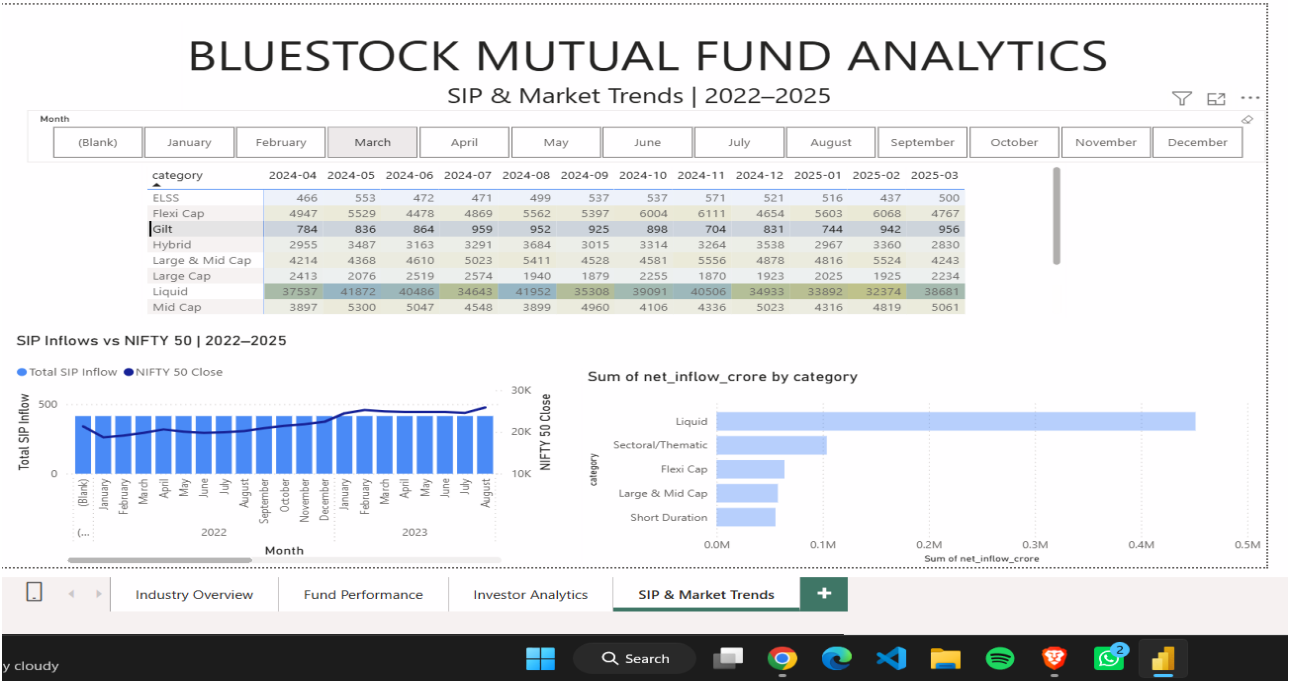
The executive page combines headline AUM, SIP , folio and scheme KPIs with AUM trend and fund-house comparison.

11. Power BI – Fund Performance

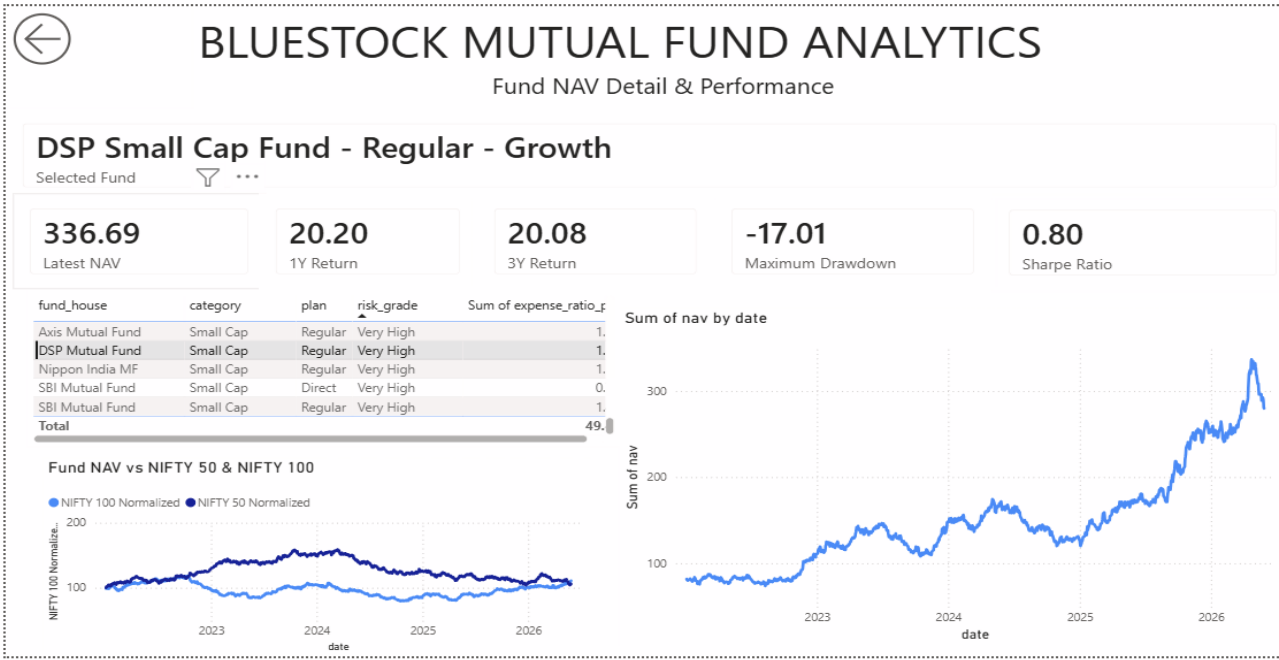


This page combines fund-house, category and plan slicers with a return-versus-risk scatter plot, fund scorecard and normalized NAV/benchmark comparison.

12. Power BI – SIP & Market Trends



13. Power BI – Fund NAV Detail



The drill-through page provides selected-fund KPIs, scheme information, NAV history and benchmark comparison.

14. Advanced Analytics – Fund Recommender

The recommender is a transparent weighted scoring framework built from the performance metrics rather than a black-box model.

Risk profile	Scoring emphasis
Conservative	Higher weight on Sharpe, volatility and drawdown
Moderate	Balanced return and risk-adjusted performance
Aggressive	Higher weight on return and market exposure

Scoring dimensions

- 3-year return/CAGR.
- Sharpe Ratio.
- Annualized volatility, with lower volatility scoring better.
- Maximum drawdown, with smaller downside scoring better.
- Beta proximity to 1.

The output ranks funds and assigns transparent labels such as Strong Buy, Buy, Hold, Watch and Avoid. The recommender is an analytical ranking component for the project and is not personalized investment advice.

15. Key Findings

Industry scale: Selected fund-house AUM reaches 62.74 lakh crore at the latest observation.

SIP momentum: Monthly SIP inflow reaches ■31,002 crore in the latest observation versus ■11,517 crore at the start of the series.

Investor participation: The latest folio count is 26.12 crore.

Return dispersion: 3-year scheme returns range from 5.14% to 23.39%.

Category differences: Small Cap has the highest average 3-year return among supplied categories at 21.69%.

Risk-adjusted performance: The highest reported Sharpe Ratio is 7.68.

Investor flows: Lumpsum transactions represent the largest transaction-value component in the supplied investor sample.

Benchmark breadth: Seven benchmark indices support comparisons across equity, mid-cap, small-cap and fixed-income-oriented segments.

16. Limitations and Future Scope

Limitations

- The analysis uses a selected 40-scheme universe rather than the complete mutual-fund market.
- Historical performance is not a guarantee of future performance.
- Historical VaR depends on the observed return distribution and may not capture extreme future events.
- The recommendation model is rule-based and does not incorporate complete investor suitability information.

Future scope

- Automated scheduled NAV ingestion and ETL execution.
- Monte Carlo NAV forecasting with uncertainty bands.
- Markowitz Efficient Frontier portfolio optimization.
- Streamlit web application.
- Automated weekly HTML/email performance reporting.

17. Conclusion and Deliverables

The Bluestock Mutual Fund Analytics capstone demonstrates an end-to-end workflow that connects raw financial data with quantitative analysis and interactive business reporting. The solution integrates data engineering, SQL, financial statistics and Power BI into one coherent analytical product.

The resulting dashboard and analytical outputs provide a structured way to explore fund performance, risk, investor behaviour, category flows and benchmark relationships.

Core deliverables

ID	Deliverable	Project output
D1	ETL Pipeline	Python ingestion, transformation and validation
D2	SQLite Database	Relational storage and SQL analysis
D3	EDA Notebook	Exploration, charts and documented insights
D4	Performance Metrics	CAGR, Sharpe, Beta, VaR and risk metrics
D5	Interactive Dashboard	Power BI multi-page dashboard and drill-through
D6	Advanced Analytics	Fund recommendation scoring
D7	Final Report	This report

End of Report